



▲ GLASSWORT

Looking like a tiny, spineless cactus, this plant lives in the wettest part of the salt marsh, flooded at every high tide.

PIONEER PLANTS

The first plants to take root in the tidal mudflats are cordgrass and leafless, juicy-stemmed plants such as glasswort. These plants can cope with being submerged by tidal saltwater twice a day, every day, and they have special adaptations to deal with the salt. Their roots bind the mud together. They also trap more particles from the water, slowly raising the level of the mudflat.

MARSH ZONES

Over time, the pioneer plants raise the mud level, so the lower marsh becomes drier and less salty. Different plants, such as sea lavender, take root in this higher zone, trapping sediment and raising the level even further. At the top of the marsh, the salt marsh plants give way to freshwater plants.

► FLOODED MARSH

High spring tides like this one flood the entire marsh with salty water. But during neap tides, only the lowest levels are flooded.

Salt marshes

The tidal shores of estuaries and deltas are too salty for most plant life. But a few specialized plants are able to cope with the salt, and even survive being flooded by saltwater at high tide. In the cooler parts of the world, these specialists are the grasses and other low-growing plants that form salt marshes. Dotted with pools and muddy creeks, the marshes provide safe refuges for many types of coastal wildlife.

QUIET LAGOONS

Salt marsh plants usually take root in quiet estuaries and lagoons that are cut off from the sea by sandy spits and islands. These natural barriers shelter the plants from the waves that might uproot them. The still water also allows fine mud to settle and build up, so it can support more and more plants. In time, the salt marsh may take over the whole area, aside from a central river channel bordered by bare mudflats.

KEY

- Salt marshes
- Mudflats

